

WORKSHEET 2: HYPOTHESIS TESTS

Instructions: This worksheet presents three examples of hypothesis tests. Please work in groups. You will be asked to turn in your group worksheet on April 10.

Background on chi-squared statistic

Suppose we have a multinomial random variable with k outcomes, labelled $1, \dots, k$, constructed from n trials such that in each trial outcome i occurs with probability p_i . Recall, the *chi-squared statistic* is defined as

$$\chi^2 := \sum_{i=1}^k \frac{(N_i - E_i)^2}{E_i},$$

where $E_i =$ (expected # of times outcome i occurs), and $N_i =$ (# of times outcome i is observed).

We'll illustrate the use of this statistic in two different types of hypothesis tests:

- For a “goodness of fit test”, χ^2 has a chi-squared distribution with $k - 1$ degrees of freedom.
- For a “test of independence” using an $r \times c$ contingency table, χ^2 has a chi-squared distribution with $(r - 1) \times (c - 1)$ degrees of freedom.

Recall, the pdf of a chi-squared random variable with d degrees of freedom is

$$f(x) = \begin{cases} \frac{1}{2^{d/2}\Gamma(d/2)} x^{\frac{d}{2}-1} e^{-x/2} & : x \geq 0 \\ 0 & : x < 0 \end{cases}.$$

0.1. Hypothesis test type #1: A “goodness of fit” test.

Problem 1 (Goodness-of-fit test). According to the *Mars Wrigley Confectionery Company*, the five colors of skittles are produced in equal proportions. Is that really true?

Let's take their claim to be the null hypothesis.

$$H_0 : p_{\text{green}} = p_{\text{yellow}} = p_{\text{red}} = p_{\text{orange}} = p_{\text{purple}} = 0.2.$$

The alternative hypothesis is

$$H_a : \text{at least one of the probabilities is different from 0.2}$$

- (a) Go grab a random sample of at least 30 skittles. What is your sample size?
- (b) For each $i \in \{\text{green, yellow, red, orange, purple}\}$, let E_i denote the expected number of skittles of color i in your sample, assuming that H_0 is true. Find E_i for all i .
- (c) Fill in the following table with expected and observed counts:

	green	yellow	red	orange	purple
expected					
observed					

- (d) Compute the chi-squared statistic χ^2 for your sample of skittles.
- (e) Circle the correct phrase to complete the sentence: *A large chi-square value means the observed data is (very different from / very similar to) the expected data.*
- (f) Under the null hypothesis of a goodness-of-fit test, χ^2 has a chi-squared distribution with $k - 1$ degrees of freedom. Set up and evaluate an integral to determine the p-value $\mathbb{P}[\chi^2 \geq u]$, where u is the value you got in part (d).
- (g) Write a sentence or two interpreting your result.

0.2. Hypothesis test type #1: A one-sided test of parameter value.

Problem 2. The organization *Physicians for Human Rights* documented attacks on health care facilities during the Syrian civil war. During the period from January 2012 through September 30, 2015, the average rate of such attacks by Syrian government forces was approximately

$$\theta_0 = 76 \text{ attacks per year.}$$

On September 30, 2015, the Russian military began an air campaign in support of the Syrian government, substantially increasing aerial operations. From that date until December 31, 2016 (a period of $t = 1.25$ years), there were $X = 146$ documented attacks on healthcare facilities by Syrian and Russian forces.

- (a) Let θ denote the rate of attacks per year during the time period after September 30, 2015. Suppose we wish to assess whether Russian intervention coincided with an increase in the rate of attacks on healthcare facilities, or whether the rate of such attacks stayed the same. State the relevant null and alternative hypotheses.

- (b) Let $\hat{\theta} = X/t$ be the observed rate of attacks in the later period, and define the test statistic

$$T = \frac{\hat{\theta} - \theta_0}{\sqrt{\theta_0/t}}.$$

Make the following two assumptions:

- Assume that if the rate of attacks per year is θ , then the number of attacks in a period of t years is Poisson distributed with mean $\lambda = t\theta$; i.e., so that $X \sim \text{Poisson}(t\theta)$.
- Assume that if $Y \sim \text{Poisson}(\lambda)$ and $\lambda > 5$ then Y is approximately normally distributed with mean λ and variance λ .

Using these two assumptions, show that T has approximately a standard normal distribution if H_0 is true.

- (c) Using T as a test statistic, formulate a rejection region of the form $[T > k]$ at a significance level of $\alpha = .05$. In other words, find the value of k such that

$$\mathbb{P}[T > k \mid H_0 \text{ is true}] = .05$$

Sketch a picture to illustrate your choice of k .

- (d) Compute the observed value of T and decide whether to reject H_0 or not.
- (e) Compute the corresponding p -value for your test statistic. (Recall, the p -value is the probability of observing a value at least as large as what you observed, assuming H_0 is true.)

0.3. Hypothesis test type #3: Test of independence.

Problem 3 (Test of Independence). A 2007 study of $n = 186$ autistic Japanese children examined a possible relationship between the onset of autism-related developmental regression and exposure to the measles mumps rubella (MMR) vaccine.¹ One of their analyses involved the following dataset:

Table of observed values

	Vaccinated	Not Vaccinated
Regression	15	39
No regression	45	87

We'll consider the following hypothesis testing problem:

H_0 : vaccination status and autistic regression are independent
versus

H_a : there is a relationship between vaccination status and autistic regression

(a) From the table of observed values, find:

- the total number of children exhibiting autistic regression
- the total number of children not exhibiting autistic regression
- the total number of vaccinated children
- the total number of unvaccinated children
- the total number of children

(b) Let p be the probability that a child exhibits autistic regression, and let q be the probability that a child is vaccinated. Use your answers from the previous part to estimate p and q .

(c) If H_0 is true, then expected number of children who are both vaccinated and exhibit autistic regression is $e_{11} = npq$. Find e_{11}, e_{12}, e_{21} , and e_{22} , and arrange them in the following table:

Table of expected counts

	Vaccinated	Not Vaccinated
Regression	$e_{11} =$	$e_{12} =$
No regression	$e_{21} =$	$e_{22} =$

(d) Compute the chi-squared statistic

$$\chi^2 = \sum_{\text{all } i,j} \frac{(o_{ij} - e_{ij})^2}{e_{ij}}$$

where o_{11}, o_{12}, o_{21} , and o_{22} are the observed values.

(e) Set up and evaluate an integral to compute a p -value for your χ^2 statistic.

(f) Write a sentence or two interpreting your result.

¹Uchiyama et al, *J Autism Dev Disord*, 2007. <https://doi.org/10.1007/s10803-006-0157-3>

Problem 4 (Simpson’s Paradox: The Perils of Aggregation). In this problem we compare the performance of two medical centers, Hospital A and Hospital B. We have the following survival data for surgery patients from these two hospitals from the last 6 weeks:

Outcome	Hospital A	Hospital B
Died	63	16
Survived	2037	784

- (a) Compare the survival rates of the two hospitals: what percentage of patients at Hospital A survive? What about Hospital B?
- (b) Is there a relationship between patient survival and which hospital they go to? Formulate and perform a chi-squared test of independence. What p -value do you get? In one or two sentences, state your null hypothesis and draw a conclusion about it.
- (c) We now consider a third factor: the condition of the patient when they are admitted for surgery. Patients are either classified as “poor condition” or “good condition”. Here’s the more detailed data:

Good Condition			Poor Condition		
Outcome	Hospital A	Hospital B	Outcome	Hospital A	Hospital B
Died	6	8	Died	57	8
Survived	594	592	Survived	1443	192

What fraction of Hospital A’s patients were classified as good condition? What about Hospital B?

- (d) Out of the patients in **good condition** who went to hospital A, what fraction survived? What about hospital B? Compare these two survival rates.
- (e) Out of the patients in **bad condition** who went to hospital A, what fraction survived? What about hospital B? Compare these two survival rates.
- (f) Compare your answers to parts (a) (d) (e). What you are observing is called *Simpson’s Paradox*. How can this discrepancy be explained? If you are facing surgery, should you go to Hospital A or Hospital B?
- (g) Fill in the following contingency table:

Outcome	Good Condition	Bad Condition
Died		
Survived		

- (h) Is there a relationship between patient survival and the condition that they are admitted for surgery? Formulate and perform a chi-squared test of independence and report a p -value.

Problem 5 (Another example of Simpson's Paradox). The influence of race on imposition of the death penalty for murder has been much studied and contested in the courts. The following three-way table classifies 326 cases in which the defendant was convicted of murder.² The three variables are the defendant's race, the victim's race, and whether or not the defendant was sentenced to death.

White Defendants			Black Defendants		
Victim Race	Yes	No	Victim Race	Yes	No
White	19	132	White	11	52
Black	0	9	Black	6	97

- (a) From these data make a 2-way table of defendant's race by death penalty. Does it look like there is a relationship between these two things?
- (b) Suppose you performed a chi-squared test of independence and for the table you just made. You get $p = .64$. How do you interpret this result?
- (c) Show that Simpson's paradox holds: specifically, that
- a higher percent of white defendants are sentenced to death overall
 - but for both black and white victims, a higher percent of black defendants are sentenced to death.
- (d) Basing your reasoning on the data, explain why the paradox holds in language that a judge could understand.

²Data from M. Radelet, *American Sociological Review*, 1981. <https://doi.org/10.2307/2095088>